

Amit Weis

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Education

University of Waterloo

Bachelor of Applied Science in Mechatronics Engineering

2025 - Present

Cumulative Average: 92.68%

Summary Of Skills

DISCIPLINES

Robotics, Software Dev, Comp Sci, Geodesy, Drones

COMMUNICATION

Leadership, Public Speaking, Budget Mgmt, Project Mgmt

SOFTWARE

C++, Python, ROS, Linux, Isaac Sim/Lab, Unity, Git

HARDWARE

Raspi, Arduino, ESP32, KiCad, SolidWorks, AutoCAD

Experience

Link Horizon

Field Drone Operations Intern

May 2026 - Present

Ottawa, Ontario

- Improved our mapping data to be **centimeter-accurate** saving our client over **\$200,000**.
- Developed the **photogrammetry** pipeline to integrate aerial datasets into Link Horizon's **AI-driven** software platform
- Managed the drone operations from pre-mission planning using **Jira**, to flight and post-processing using the DJI ecosystem.

University of Waterloo Intelligent Vehicle Organization

Co-Founder

May 2026 - Present

Waterloo, Ontario

- Co-founding a official student design team in partnership with **Waterloo RoboHub** to design and build autonomous vehicle platforms while managing a growing team of almost **a dozen students**.
- Researching **SLAM** to integrate into the team's first project, an **autonomous submarine** for the RoboSub competition.
- Simulating the autonomous submarine's control system in **Stonefish**, a physics accurate **ROS** library

University of Waterloo Biomechatronics Design Team

Software Team Member

Jan 2026 - May 2026

Waterloo, Ontario

- Built a **human musculoskeletal simulation** in **Isaac Sim** for a soft exoskeleton to follow, to provide a foundation to train the **reinforcement learning** control policy for the exosuit in **Isaac Lab**.
- Designing a movement-assisting algorithm using **Kalman filtering** to predict and adapt to user intent in real time.

Packet39

Software Developer

Jan 2023 - May 2026

Waterloo, Ontario

- Optimized a **custom augmented reality engine** that combined machine learning and computer vision by contributing and annotating data for the Rod Laver Tennis Arena in Melbourne, Australia.
- Optimized Materials for an augmented reality exposure therapy app for PTSD patients at the Wayne State University hospital that won the "**XR Healthcare Solution of the Year**" at the AIXR EuropeXR Awards 2025.
- Developed an **Arduino** capacitive touch and hover detection system for an interactive artifact wall at AbbVie's Allergan Medical Institute training centers in Orange County and Atlanta, deployed across a medical device history exhibit.

Projects

see [website](#) for more info

Autonomous Pool Table - Bttrpie

- Built a mini-pool table with an **autonomous ball sorting system** using custom 3D printed components, machined aluminum parts using a **Stellaris Cortex-M4F microcontroller** for computation and signal control.
- Implemented **PID motion control** for a rack-and-pinion gantry that has accurate odometry for blind positioning.
- Using **SolidWorks**, designed all mechanical components and validated the physical design through structural simulations.

Companion Robot - aRTy

- Designed a companion robot in collaboration with **Alex Loker** and **Elevait Labs**, integrating their proprietary AI system onto a Raspberry Pi 5 in a custom 3D-Printed shell to enable natural human-robot interaction.
- Pre-rendered facial expressions and emotions for mounted LCD screen to ensure anthropomorphic interaction

Leadership

MechMania

Co-Lead

May 2024 - May 2025

Waterloo, Ontario

- Led a **team of 10+** through all stages of event planning, logistics, and execution from conception to completion.
- Founded a flagship high school robotics competition sponsored by Waterloo Robohub, SolidWorks, and Bittele now entering its **fourth consecutive year** with **120+ competing students** every year.
- Secured **thousands of dollars** in funding through targeted sponsor outreach and strategic networking.

University of Waterloo Engineering Society

Academic Representative

Sep 2025 - Present

Waterloo, Ontario

- Represented **100+ students** in all academic and society matters, including grade petition facilitation, examination fairness disputes, and course rescheduling requests.